GGG-T-80A
November 24, 1970
SUPERSEDING
Fed. Spec. GGG-T-80
September 25, 1956
(see 6.3)

FEDERAL SPECIFICATION

# TAPS, THREAD CUTTING, CARBON STEEL HAND, CUT THREAD, AMERICAN STANDARD TAPER PIPE THREADS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers carbon steel, taper pipe taps having cut threads, conforming to the American Standard pipe form.

1.1.1 Federal specification coverage. This specification does not include all types of taps indicated by the title of this specification, but is intended to cover only the types and series which are generally used by the Federal Government.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specification:
PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Port Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS) Handbook:
H28, Part II - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D. C. 20402.)

FSC 5136

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) Standards:

276.4 - Standard Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness, and Knoop Hardness) (ASTM E140).

B94.9 - Taps-Cut and Ground Threads.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is descriptive and not restrictive, and is not intended to preclude the purchase of taps otherwise conforming to this specification.

3.2 Design and construction. Taps shall be designed and constructed to meet the requirements of ANSI B94.9 for carbon steel, cut thread, taper pipe taps. The taps furnished by the supplier shall be new, and the standard current product that has already performed satisfactorily in industry.

3.3 Thread form. The American standard pipe thread form shall conform to Handbook H28, Part II.

3.4 Material. The carbon steel used in the taps shall be of such composition as to produce taps of the highest quality manufactured in the tool industry.

3.5 Marking. The taps shall be permanently marked with the manufacturer's, name or with a trade-mark of such known character that the source of manufacture may be readily determined. In addition, the taps shall be marked with their nominal sizes, and threads per inch. Taper pipe taps shall be marked with symbol "NPT".

3.6 Hand of thread. Unless otherwise specified (see 6.2), right-hand cutting taps shall be furnished.

3.7 Tap centers. The cut-thread taps shall have suitable centers provided at each end. All taps shall have internal centers on the thread end.

3.8 Finish. All surfaces of the tap shall be smooth, with the flutes and shanks ground or polished after hardening.

3.8.1 Threads. The threads (flanks and crests) shall have a smooth finish, free from scale, burrs, nicks, cutter chatter marks, or other imperfections which may affect the accuracy of the tap.

3.8.2 Flutes. The straight flutes shall be smoothly ground or polished after hardening. Flutes shall be accurately indexed and of sufficient depth to provide ample chip clearance.

3.9 Hardness. The full length of the threaded end of the tap shall be suitably hardened (see 4.3.1) and tempered. The taps shall have a Rockwell hardness of not less than 61 nor more than 65 on the C scale when measured in the first 80 percent of the threaded section. The shanks and squares shall have a minimum Rockwell hardness of C-45 to prevent either twisting or breaking.

3.10 Eccentricity tolerances. Eccentricity tolerances of tap elements for cut threads shall be as specified in table I.



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TABLE I. Eccentricity tolerances of tap elements for cut threads

| Element | Range Sizes are Inclusive: Hand Mch. Screw | Range Sizes are Inclusive: Pipe | Cut Thread: Eccentricity | Cut Thread: t.i.v. 1/ |
|---|---|---|---|---|
| | Inches | Inches | Inch | Inch |
| Square (at central point) | 0-1/2 | 1/16-1/8 | .0030 | .0060 |
| | 17/32-4 | 1/4-4 | .0040 | .0080 |
| Shank | 0-5/16 | 1/16 | .0030 | .0060 |
| | 11/32-4 | 1/8-4 | .0040 | .0080 |
| Major Diameter | 0-5/16 | 1/16 | .0025 | .0050 |
| | 11/32-4 | 1/8-4 | .0040 | .0080 |
| Pitch Diameter (at first full thread) | 0-5/16 | 1/16 | .0025 | .0050 |
| | 11/32-4 | 1/8-4 | .0040 | .0080 |
| Chamfer 2/ | 0-1/2 | 1/16-1/8 | .0020 | .0040 |
| | 17/32-4 | 1/4-4 | .0030 | .0060 |

1/ t.i.v. = Total indicator variation. Figures are given for both eccentricity and total indicator variation to avoid misunderstanding.
2/ Chamfer should preferably be inspected by light projection to avoid errors due to indicator contact points dropping into the thread grooves.
Note: Metric taps shall conform to the nearest inch size above.

3.11 Chamfers. Chamfers shall be smooth and have a minimum of chatter marks.

3.12 Dimensions and tolerances. Dimensions of taps shall conform to table II. Tolerances shall conform to table III for each specified size (see 6.2).

3.12.1 Lead tolerance. A maximum lead error of plus or minus 0.003 inch in 1 inch of thread is permitted.

TABLE II. Dimensions for American Standard taper pipe thread taps

| Nominal size | Dimensions: Length, overall | Dimensions: Length of thread | Dimensions: Length of square | Dimensions: Diameter of shank | Dimensions: Size of square | Number of flutes |
|---|---|---|---|---|---|---|
| Inches | Inches | Inches | Inches | Inches | Inches | Inches |
| 1/8 | 2-1/8 | 3/4 | 3/8 | 0.3125 | 0.234 | 4 |
| 1/8 | 2-1/8 | 3/4 | 3/8 | .4375 | .328 | 4 |
| 1/4 | 2-7/16 | 1-1/16 | 7/16 | .5625 | .421 | 4 |
| 3/8 | 2-9/16 | 1-1/16 | 1/2 | .7000 | .531 | 4 |
| 1/2 | 3-1/8 | 1-3/8 | 5/8 | .8875 | .515 | 4 |
| 3/4 | 3-1/4 | 1-3/8 | 11/16 | .9063 | .679 | 5 |
| 1 | 3-3/4 | 1-3/4 | 13/16 | 1.1250 | .843 | 5 |
| 1-1/4 | 4 | 1-3/4 | 15/16 | 1.3125 | .984 | 5 |
| 1-1/2 | 4-1/4 | 1-3/4 | 1 | 1.5000 | 1.125 | 7 |
| 2 | 4-1/2 | 1-3/4 | 1-1/8 | 1.8750 | 1.406 | 7 |
| 2-1/2 | 5-1/2 | 2-9/16 | 1-1/4 | 2.2500 | 1.687 | 8 or 9 |
| 3 | 6 | 2-5/8 | 1-3/8 | 2.6250 | 1.968 | 8 or 9 |

TABLE III. Tolerances of dimensions

| | Range | Tolerance |
|---|---|---|
| | Inches | Inch |
| Length, overall ---- | 1/8 to 3/4 incl. | ± 1/32 |
| | 1 to 3 incl. | ± 1/16 |
| Length of thread ---- | 1/8 to 3/4 incl. | ± 1/16 |
| | 1 to 1-1/4 incl. | ± 3/32 |
| | 1-1/2 to 3 incl. | ± 1/8 |
| Length of square ---- | 1/8 to 3/4 incl. | ± 1/32 |
| | 1 to 3 incl. | ± 1/16 |
| Diameter of shank --- | 1/8 to 1/2 incl. | -0.0070 |
| | 3/4 to 3 incl. | -0.0090 |
| Size of square ------ | 1/8 | -0.0040 |
| | 1/4 to 3/4 incl. | -0.0060 |
| | 1 to 3 incl. | -0.0080 |



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3.12.2 Angle tolerance. The permissible errors for half and full angles shall be as shown in table IV.

TABLE IV. Angle tolerance

| Threads per inch | Angle tolerance | |
|---|---|---|
| | Error in half angle | Error in full angle |
| | Minutes | Minutes |
| 8 | ± 40 | ± 60 |
| 11-1/2 to 27 incl. | ± 45 | ± 68 |

3.13 Taper pipe tap. The taper pipe taps shall be similar to figure 1.

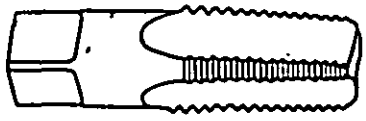


Figure 1 - Taper pipe tap.

3.13.1 Thread clearance. Taper pipe tap threads shall have radial relief.

3.13.2 Chamfering. The tapered pipe threads shall be chamfered on the starting end for a distance of from two to three and one half threads.

3.13.3 Thread limits and tolerances. Tapered taps shall conform to the gage measurements as shown in table V and to the tolerances shown in table VI.

3.13.4 Taper of pipe taps. All tapered pipe threading taps shall produce threads tapered 3/4 inch to the foot governed by the taper per foot tolerances shown in table VI.

TABLE V. Thread limits of taper pipe taps

| Nominal size 1/ | Threads per inch | Gage 1/ measurement |
|---|---|---|
| Inches | | Projection inch |
| 1/8 | 27 | .312 |
| 1/4 | 18 | .459 |
| 3/8 | 18 | .454 |
| 1/2 | 14 | .579 |
| 3/4 | 14 | .565 |
| 1 | 11-1/2 | .678 |
| 1-1/4 | 11-1/2 | .686 |
| 1-1/2 | 11-1/2 | .699 |
| 2 | 11-1/2 | .667 |
| 2-1/2 | 8 | .925 |
| 3 | 8 | .925 |

1/ Distance small end of tap projects through American Standard pipe thread ring gage.

TABLE VI. Tolerances for gage measurement projection and taper per foot of thread

| | Tap size | Tolerance | |
|---|---|---|---|
| Gage measurement projection | Inches<br>1/8 to 3/4 incl.<br>1 to 3 incl. | Inch<br>± 1/16<br>± 3/32 | |
| | | Min. | Max. |
| Taper per foot of thread | 1/8 to 3/8 incl. | Inch<br>23/32 | Inch<br>27/32 |
| | 1/2 to 2 incl. | 23/32 | 13/16 |
| | 2-1/2 to 3 incl. | 47/64 | 51/64 |

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3.14 Workmanship. All taps shall produce neat, precise, clean threads, be free from imperfections, and manufactured in accordance with the highest level of existing standards practiced in industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality assurance shall be provided through a quality conformance inspection. The inspection consists of a hardness test, an eccentricity test, a performance test, an examination, and an examination of the preparation of the shipment for delivery. Each sample tap selected shall be inspected for compliance with this specification and MIL-STD-105. The lot represented by the sample shall be rejected when the number of defectives equals or exceeds the rejection number specified in MIL-STD-105 for General Inspection Level II for a given AQL. Nonconforming taps shall not be offered for delivery.

4.2.1 Sampling for the quality conformance inspection.

4.2.1.1 Lot. A lot shall consist of all the taps of the same size, and make that are offered for delivery at one time.

4.2.1.2 Sample. The random sample of taps offered for quality conformance inspection shall be selected from each lot in accordance with the procedures shown in MIL-STD-105 for the General Inspection Level II.

4.2.2 Examination. Each of the sample taps selected in accordance with 4.2.1.2 shall be examined for compliance with the requirements specified in this specification for design, workmanship, and marking. All dimensions shall be within the tolerances specified. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.2.3 Hardness Conformance. Each of the sample taps selected in accordance with 4.2.1.2 shall be tested in accordance with 4.3.1 to determine compliance with the hardness requirements specified in 3.9. The AQL shall be 1.5 percent defective.

4.2.4 Performance. Each sample tap shall be subjected to performance tests at the discretion of the Government. A tap will be defective when it fails to pass the performance test described in 4.3.2. The AQL shall be 1.5 percent defective.

4.2.5 Examination of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5, or as specified in the contract or purchase order.

4.3 Test procedures. Each sample tap shall be tested for hardness and performance according to the test procedures delineated in paragraphs 4.3.1 and 4.3.2.

4.3.1 Hardness test. The shank and thread of the tap shall be tested for proper hardness by using extra fine, Swiss pattern, pillar testing files, cut Nos. 0 and 1, having the proper Rockwell C hardness number. Each tap to be tested shall be mounted in a clamping device. Then, at a predetermined location on the tap, the test file will be applied using short, firm strokes. The test file used shall be in good condition. The tap shall be judged as file hard when the file glides over the tap surface without penetrating the tap surface. When the file bites into the tap surface, it shall be judged as file soft. If for any reason it is decided that a more accurate hardness reading is required, the hardness determination shall be made with appropriate hardness testing equipment with the readings converted, if necessary, to Rockwell "C" scale in accordance with ANSI Z76.4 (ASTM E-140).

4.3.1.1 Threaded portion. The threaded portion of the tap shall be tested for Rockwell C 61 - 65 hardness by applying a file of Rockwell C 60 to the heel of the land. If this portion of the tap tests soft to the file, then the tap shall be defective because it does not comply with the hardness requirements specified in 3.9. A file of Rockwell C 67 shall be applied to the heel of the land of the tap. If, the tap tests hard to the file, then the tap shall be defective because it does not conform to the requirements of 3.9. The AQL shall be 1.5 percent defective.



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4.3.1.2 Shank. The shank end of the land shall be tested for a minimum value of Rockwell C 45 hardness by applying a file of Rockwell C 44 to it. If this portion of the shank tests soft to the file, then the tap shall be judged defective, because it does not conform to the requirements specified in 3.9. The AQL shall be 1.5 percent defective.

4.3.2 Performance test. A relative tap drill, taper reamer and mild steel plate, approximating the thickness specified in table VII, shall be used for each size tap in test. Drill, taper, ream and tap eight holes (using a suitable lubricant) in the steel plate taken for each size tap in test. All tapped holes shall be checked visually for any deviation from what would be considered a well tapped hole. Taps used in tapping holes shall be rechecked for any excessive dulling of cutting edges, cracking, chipping of teeth or any other detrimental failure in the tap. If tapped holes appear uniform when checked with thread gage and the tap appears sharp and in good condition, it has passed the test.

TABLE VII. Tap drill sizes and mild steel plate thickness for performance tests

| Nominal size | Threads per inch | Tap drill size | Mild steel plate thickness |
|---|---|---|---|
| | | Inches | Inches |
| 1/8 | 27 | "R" | 3/8 |
| 1/4 | 18 | 7/16 | 1/2 |
| 3/8 | 18 | 37/64 | 1/2 |
| 1/2 | 14 | 45/64 | 5/8 |
| 3/4 | 14 | 59/64 | 5/8 |
| 1 | 11-1/2 | 1-5/32 | 3/4 |
| 1-1/4 | 11-1/2 | 1-1/2 | 7/8 |
| 1-1/2 | 11-1/2 | 1-47/64 | 7/8 |
| 2 | 11-1/2 | 2-7/32 | 1 |
| 2-1/2 | 8 | 2-5/8 | 1-1/4 |
| 3 | 8 | 3-1/4 | 1-1/4 |

4.3.3 Eccentricity test. The total indicator variation (TIV) shall not exceed the values listed in 3.10 when the sample taps are held on centers at their ends and rotated 360 degrees. The AQL shall be 1.5 percent defective.

4.4 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5, or as specified in the contract or purchase order.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the requirements of PPP-P-40. Levels of preservation and packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The taps covered by this specification are carbon steel, cut thread pipe taps used for cutting internal taper threads where internal pipe threads are required. These carbon steel taps are normally used for hand tapping.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see table II).
(c) If left hand threads are required (see 3.6).
(d) Selection of applicable level of preservation and packaging and the level of packing and marking (see 5.1).

6.3 Deletion. Type II straight pipe taps have been deleted from this specification. There are no stocking requirements for these taps.



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MILITARY INTEREST:

User activity:
Army - CE, GL, WC
Navy - SH, MC
Air Force - 85

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS

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